Austo Automobile Dealership

Market Analysis & Strategic Business Report

PGP-DGGA’26 Project 1: Exploratory Data Analysis

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1. Executive Summary

This report provides a comprehensive data-driven analysis of the Austo Automobile dealership's customer base and sales performance. By interrogating a dataset of 1,581 transactions, we have identified core market segments, evaluated the impact of finance and lifecycle stage of customer on revenue, and tested existing market assumptions.

Key Finding: The analysis debunked the prevalent 'Sheldon Cooper' theory that salaried males are prime targets for SUVs. In reality, this segment shows an overwhelming preference for Sedans, which represent the dealership's most consistent revenue driver across nearly all demographics.

2. Data Overview

1. Statistical Overview

A thorough diagnosis of the 1,581-record dataset was performed to ensure statistical validity.

```
df.describe(include='all').T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Age	1581.0	NaN	NaN	NaN	31.922201	8.425978	22.0	25.0	29.0	38.0	54.0
Gender	1528	4	Male	1199	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Profession	1581	2	Salaried	896	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Marital_status	1581	2	Married	1443	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Education	1581	2	Post Graduate	985	NaN	NaN	NaN	NaN	NaN	NaN	NaN
No_of_Dependents	1581.0	NaN	NaN	NaN	2.457938	0.943483	0.0	2.0	2.0	3.0	4.0
Personal_loan	1581	2	Yes	792	NaN	NaN	NaN	NaN	NaN	NaN	NaN
House_loan	1581	2	No	1054	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Partner_working	1581	2	Yes	868	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Salary	1581.0	NaN	NaN	NaN	60392.220114	14674.825044	30000.0	51900.0	59500.0	71800.0	99300.0
Partner_salary	1475.0	NaN	NaN	NaN	20225.559322	19573.149277	0.0	0.0	25600.0	38300.0	80500.0
Total_salary	1581.0	NaN	NaN	NaN	79625.996205	25545.857768	30000.0	60500.0	78000.0	95900.0	171000.0
Price	1581.0	NaN	NaN	NaN	35597.72296	13633.636545	18000.0	25000.0	31000.0	47000.0	70000.0
Make	1581	3	Sedan	702	NaN	NaN	NaN	NaN	NaN	NaN	NaN

1. This dataset contains customers in the age group (22-54), average age of buyer being ~32 and median age is 29.

2. Gender has 4 types since Female is spelt incorrectly - needs correction and also has NaN values, Male is the dominant category.

3. ~56 % of the Customers are Salaried

4. ~91 % of the Customers are Married, indicating that data is heavily skewed towards 'family' buyers who generally prefer large/practical vehicles

5. ~62 % of the Customers are Post Graduate

6. Customers typically have 2 dependents

7. ~50 % of Customers have Personal Loan indicating that this may influence their purchase decisions

8. ~67% of Customers do not have House Loan indicating that this may influence their purchase decisions

9. ~ 55 % of the Customers have working partners which may also inlfuence purchase decisions

10. Typical Salary of a Customer is 59,500 USD

11. 7 % of the data in 'Partner_salary' is missing, this is a data entry error

12. Combined Salary is typically 78,000 USD, with a significant gap between Median and Max indicating the presence of high income customers

13. Price of Vehicle is typically 31k USD, with a significant gap between Median and Max indicating some Premium priced cars

14. ~44 % of the Make present in the dataset is 'Sedan', which dominates the dataset

2. Missing Value Treatment:

1. Gender

```
Gender
Male      1199
Female     327
NaN         53
Femal        1
Femle        1
Name: count, dtype: int64
```

The Gender column had 'Female' spelled incorrectly. These were corrected under the assumption that they were simple data-entry error.

There were 53 missing values in the Gender column. These missing values were renamed to 'Unknown' to avoid introducing bias into the dataset. Since the sample size is small, this group shall be interpreted with caution.

2. Partner_salary

```
df['Partner_salary'].isnull().sum()
```

```
np.int64(106)
```

There were 106 missing values in the Partner_salary column. The missing values correspond to customers where Partner_working = Yes. These were imputed by the logic (Total_salary - Salary). This imputation was applied only to customers where Partner_working = Yes, since Total_salary represents the combined income of both partners. Applying it to single-income households would have introduced fictitious values.

3. Outlier Treatment

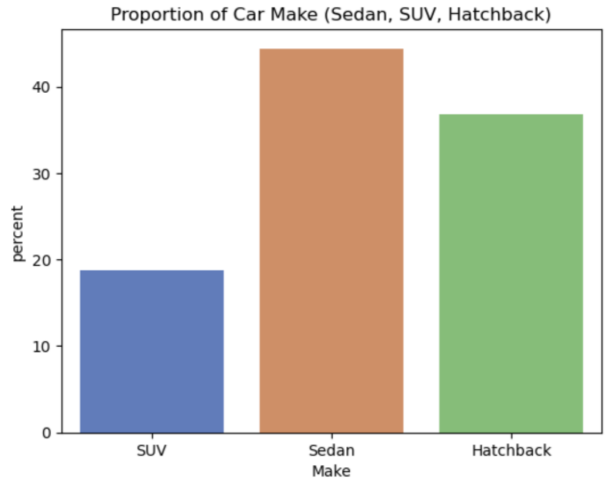
Checked for outliers in the columns Price, Salary and Total_salary.

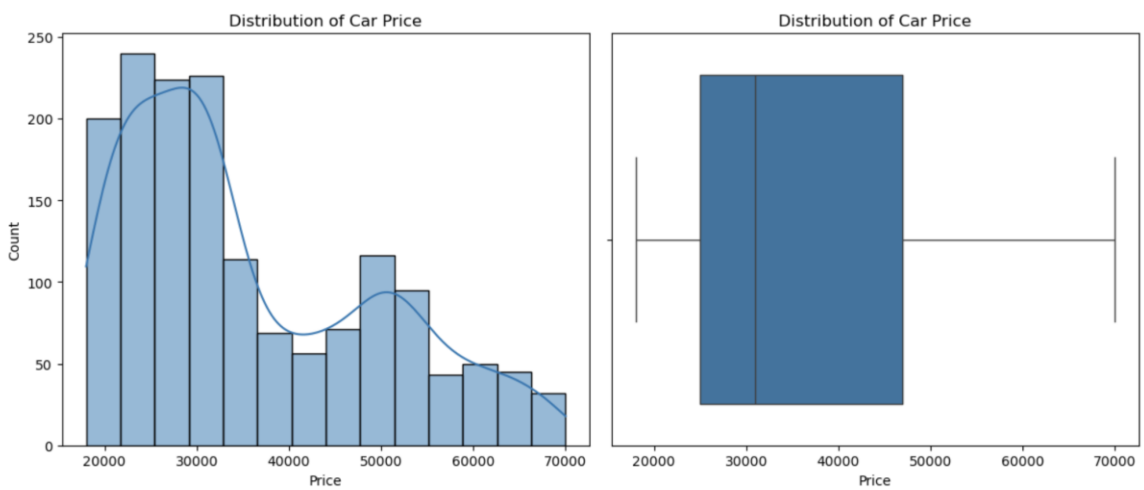
- The high price aligned to change in car price segment
- Upon inspecting the top salaries, it was found that they were legitimate salaries of customers
- The high Total_salary was combined salary of Customer and their working Partner explaining the higher combined earning capacity.

Therefore, the outliers were retained.

3. Univariate Analysis

Overview of Car Price and Car Make





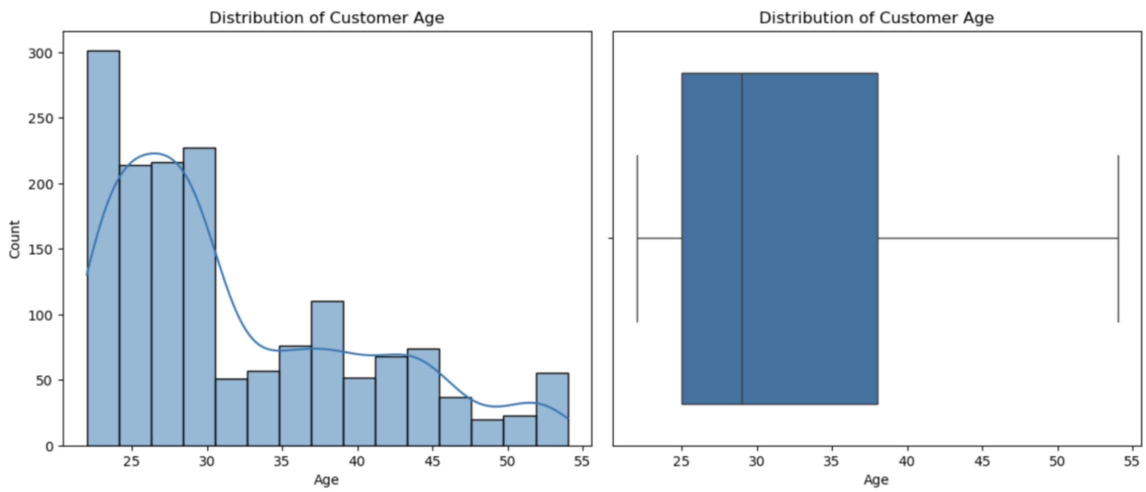
1. 50% of the cars fall in the price range 25k - 47k USD while higher priced cars are comparatively less in number stretch the long tail towards the right. This indicates that a significant portion of cars are present in the Mid segment.

2. Approximately 45% of the customers prefer Sedan in this dataset, followed by Hatchback (~ 35%) and SUV (~ 20%). This indicates that Sedans are the most purchased car type followed by Hatchback.

Understanding Customer Base

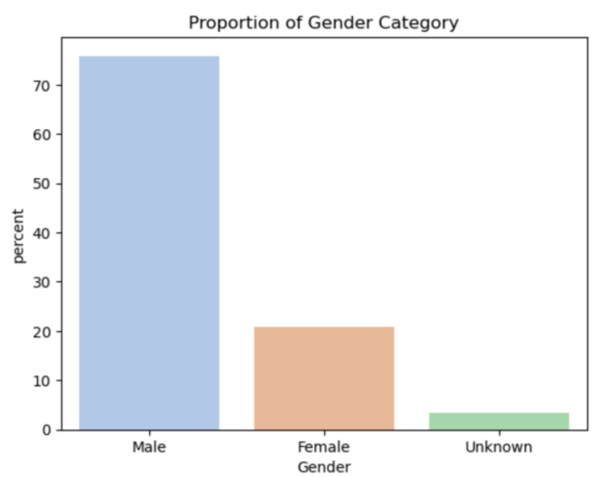
Section 1: Customer Demographics

1. Age



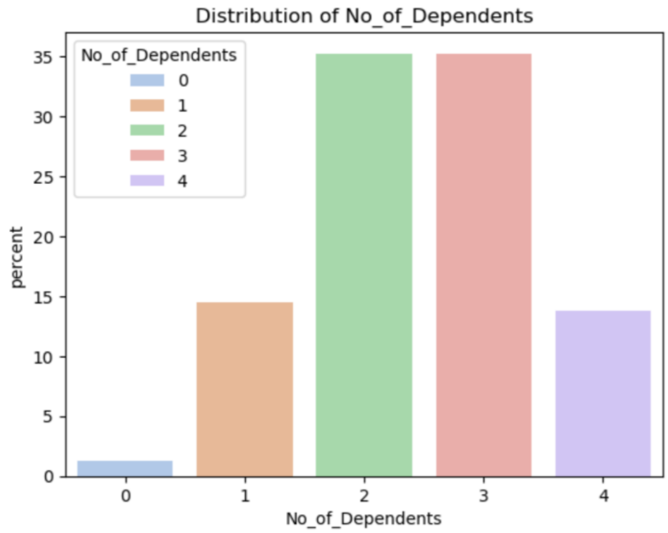
The customers in this dataset are concentrated in the age group 25-38, indicating a young customer base with potential first-time buyers or customers making family-oriented purchase decisions. The older age groups are more dispersed forming a relatively smaller sample size so interpret with caution.

2. Gender



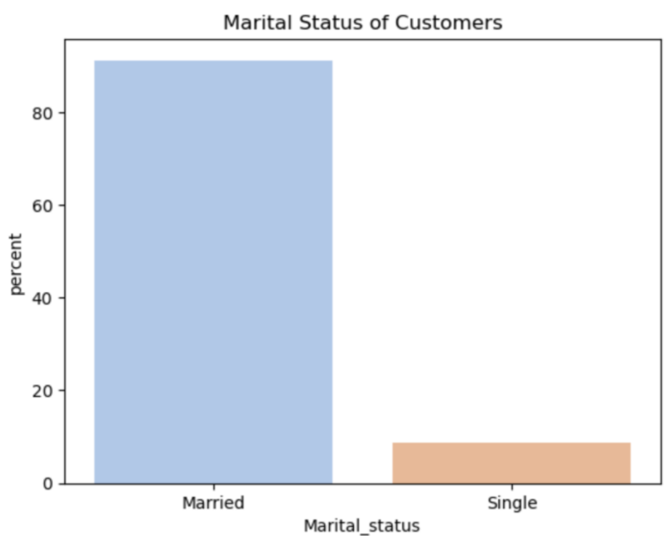
Approximately 75% of the customers are Male and 21% are Female, Unknown constitutes only 4% of the data. This dataset is heavily skewed towards male customers, therefore the insights derived will be more reflective of Male-behavior.

3. Number of Dependents



~ 70% of the customers have more than one dependent, suggesting that this dataset has a significant concentration of individuals with family responsibilities.

4. Marital Status



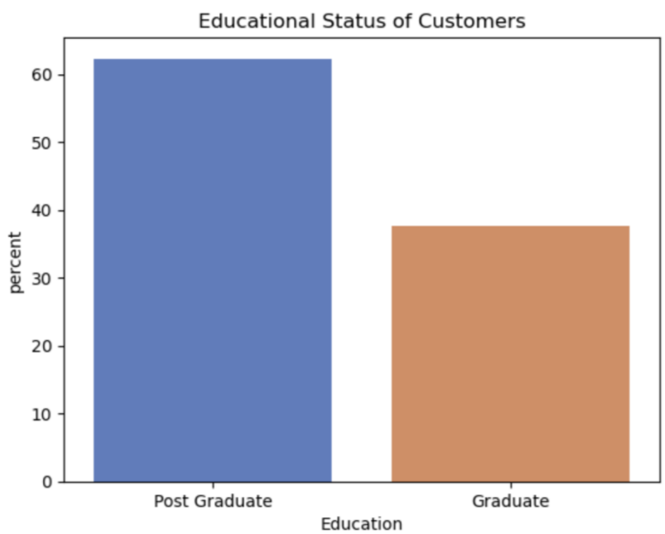
~ 85% of the customers in this dataset are married suggesting that they are more likely to make family-oriented purchases

Section 2: Socio-economic Profile

1. Profession

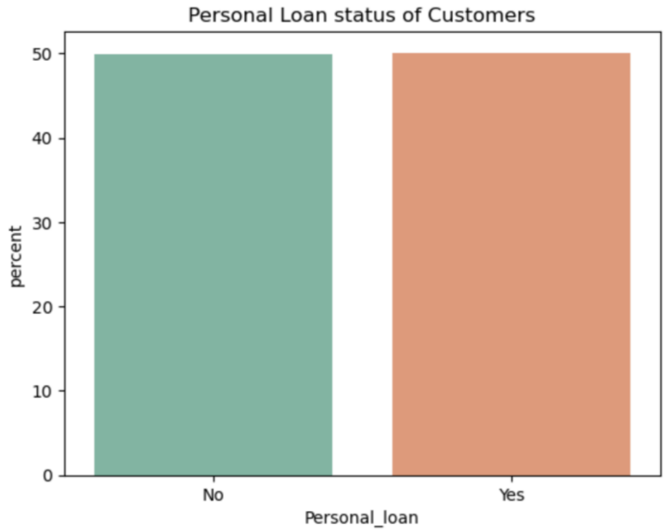
Salaried Professionals are slightly higher (~ 57 %) than business individuals (~43 %)

2. Education



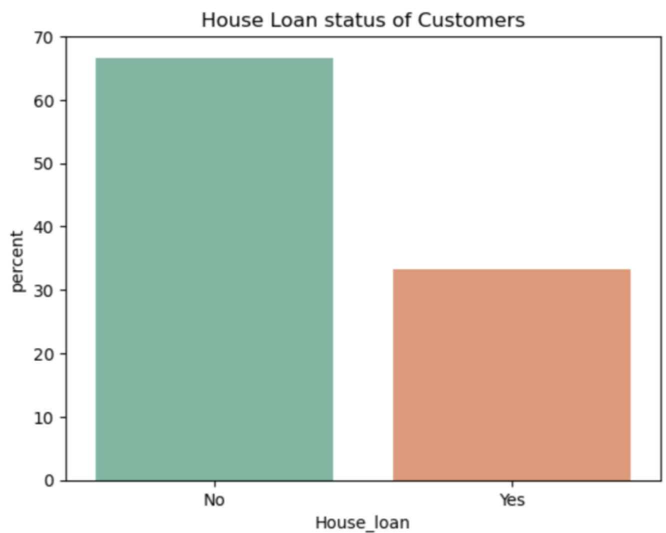
~ 63 % Customers are Postgraduates in the given dataset indicating a highly educated customer base.

3. Personal Loan Status



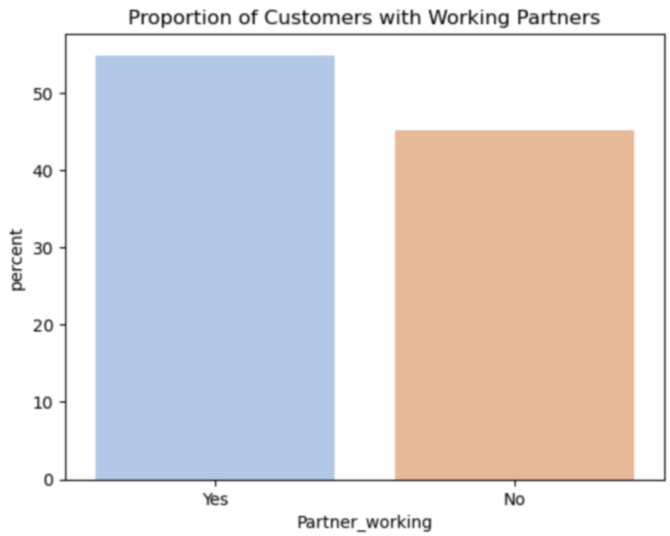
This dataset provides an equal distribution of customers with and without Personal loan providing a balanced base for comparison.

4. House Loan Status



Approximately 35% of the customers have House loans. Customers with a House Loan may have less disposable income which may influence their spending behavior.

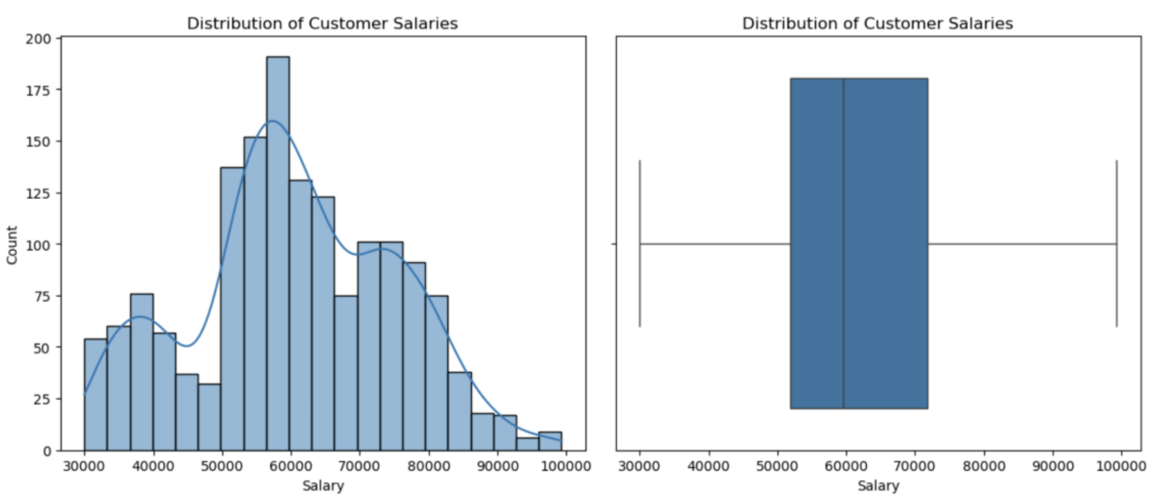
5. Partner Working Status



~ 55% of the Customers have working partners indicating the presence of Double-income households in the dataset. This suggests a higher purchasing power there by influencing the choice of Vehicles.

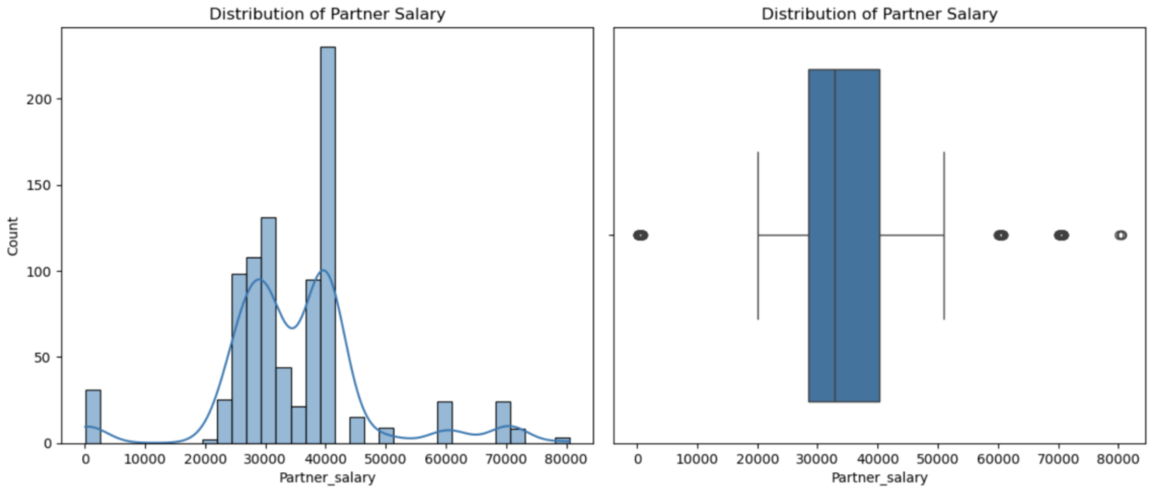
Section 3: Income Distribution

1. Customer Salary

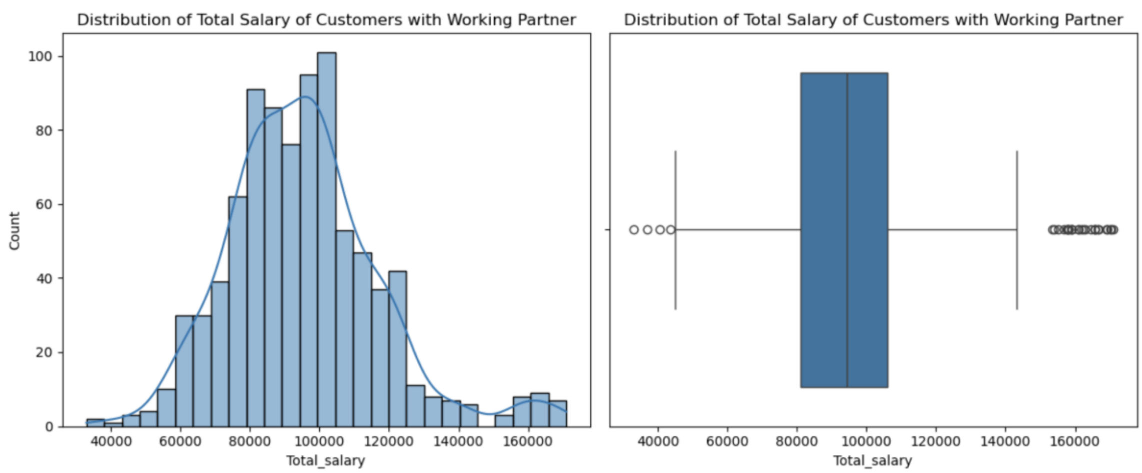


There are customers from varied salary ranges but 50% of them concentrated in the range 52k - 72k USD suggesting a mid-income range.

2. Partner Salary



3. Total Salary (Partner_working == ‘Yes’)



When considering only working partners, the salary distribution is centered around the mid-range, with a median around 32–35k USD and most values falling between approximately 28k and 40k USD. The distribution is slightly right skewed, with a few high-income outliers extending up to 80k USD.

The distribution of Total Salary among Dual-income households is right skewed, with majority of customers falling in the mid-income range. A small number of high-income households appear as outliers on the right, representing a premium segment with a much higher earning capacity.

These values are retained as they reflect a typical variation in income in an urban demographic that uses cars, rather than data anomalies.

Take-aways from Univariate Analysis

This dataset predominantly has customers who have purchased Sedans and Hatchbacks while SUVs constitute a relatively small share. A significant number of purchases were made in the Mid-price segment.

The customer base is more concentrated in the age group 25-38, and the gender distribution is more skewed towards Male. There were missing values in Gender column and has been renamed to Unknown. Even though it is a small sample size, it shall be interpreted with caution.

A large part of the customer base is highly educated, married and with dependents suggesting a family-oriented demographic. Salaried professionals form the majority compared to business customers, indicating a relatively stable income base.

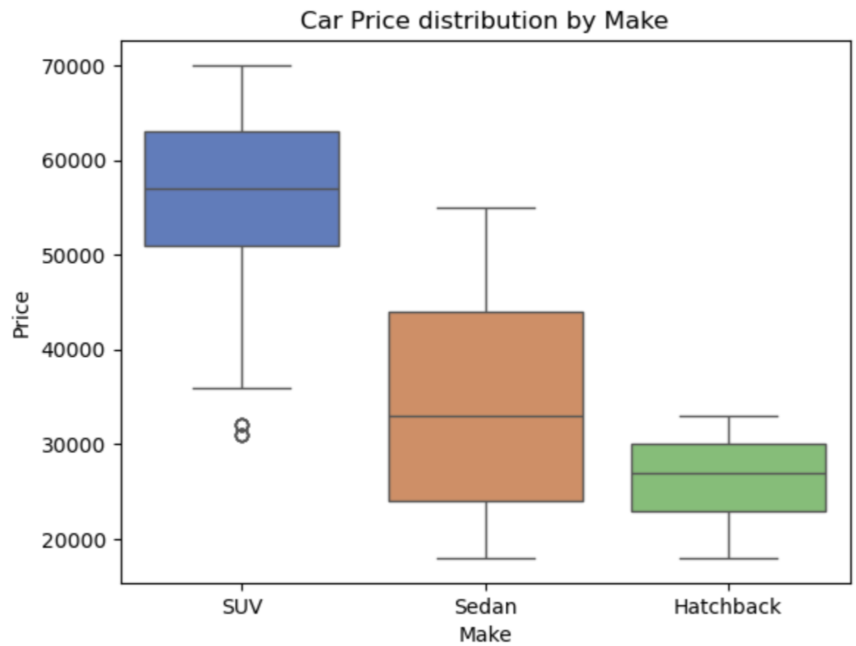
There are nearly equal number of customers with and without Personal loan providing a balanced base for analysis. Customers without home loan are higher compared to customers with home loans indicating that loan adoption is present within the customer base. Additionally, over half of the customers have working partners, which indicates the presence of dual-income households.

From a financial perspective, customers are typically distributed in a mid-income range with a small segment representing high-income households.

Overall, this dataset represents a young, urban, financially stable customer base with moderate to high income levels.

4. Bivariate Analysis

Section 1: Car Make vs Car Price

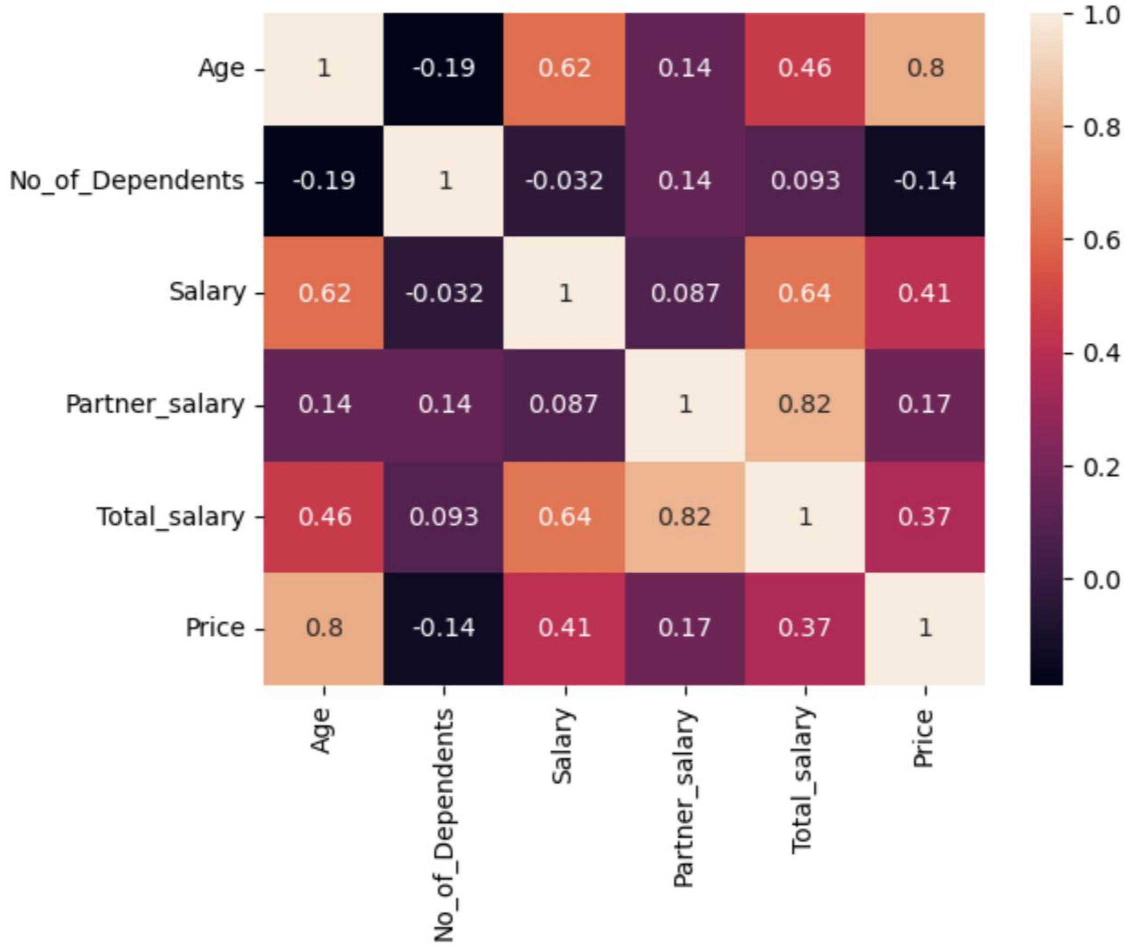


1. SUVs primarily occupy the **mid-to-premium** price segment, with higher median prices and a wider spread.

2. Sedans act as a **transitional segment**, overlapping with both hatchbacks and SUVs, while **hatchbacks** remain concentrated in the lower price range forming the **entry** segment

Section 2: Customer Attributes Vs Car Price

Correlation of all the numerical variables

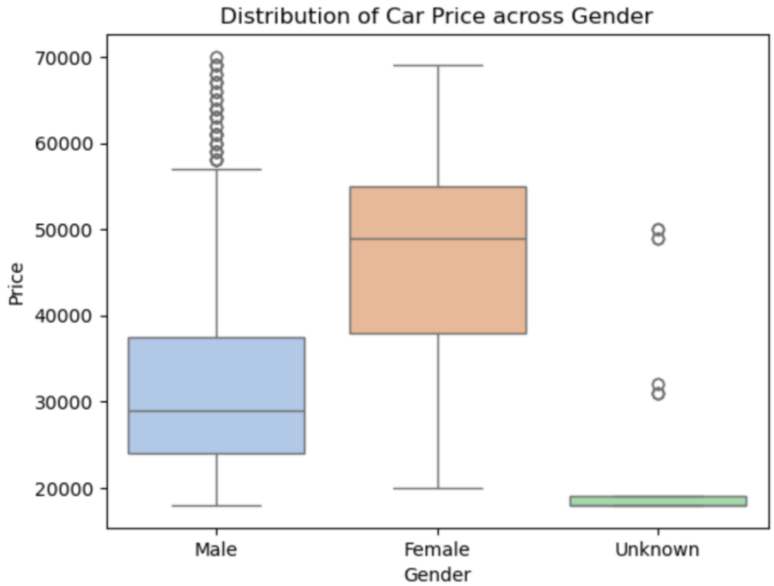


Age shows the strongest correlation with car price (~0.80), indicating that customers tend to purchase higher-priced cars as they grow older. While income also influences price (Salary ~0.41, Total Salary ~0.37), its impact is comparatively moderate. Number of Dependents show a negative correlative(-0.19) with Car Price suggesting that family responsibilities affect spending behavior.

This suggests that car purchase decisions are not driven purely by affordability, but also by life stage and evolving preferences.

Partner income shows minimal correlation (~0.17), suggesting that dual-income households do not necessarily spend more on cars.

1. Gender



Gender	count	mean	std	min	25%	50%	75%	max
Female	329.0	47705.167173	11244.836378	20000.0	38000.0	49000.0	55000.0	69000.0
Male	1199.0	32817.347790	12299.239195	18000.0	24000.0	29000.0	37500.0	70000.0
Unknown	53.0	23339.622642	10308.591257	18000.0	18000.0	18000.0	19000.0	50000.0

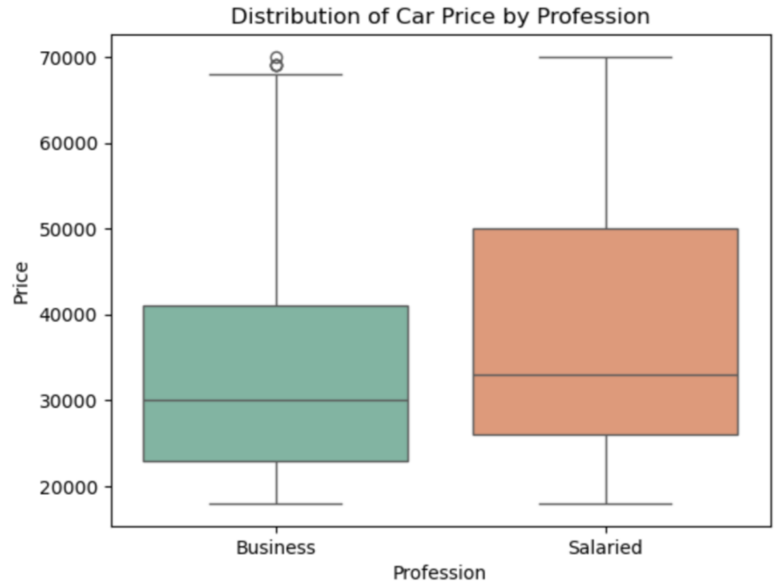
Male customers are concentrated in the low-mid price segment but contribute to a few high value outliers. Female customers are concentrated in the mid-high price segment. Typical spending levels among Female customers are higher in this dataset.

2. Age

	count	mean	std	min	25%	50%	75%	max
Age								
(21.999, 25.0]	419.0	25544.152745	4376.408365	18000.0	22000.0	25000.0	29000.0	33000.0
(25.0, 29.0]	422.0	27210.900474	7008.686476	18000.0	22000.0	26500.0	31000.0	53000.0
(29.0, 38.0]	370.0	40683.783784	11580.284468	18000.0	32000.0	40500.0	49000.0	70000.0
(38.0, 54.0]	370.0	51462.162162	10513.634234	31000.0	43250.0	52000.0	60000.0	70000.0

There is an upward shift in Price with increase in Age, which reemphasizes what was seen in the correlation heatmap. The scatterplot suggests a positive relationship between customer age and car price, with younger customers generally purchasing lower-priced vehicles and older customers showing a wider spread, including higher-priced cars.

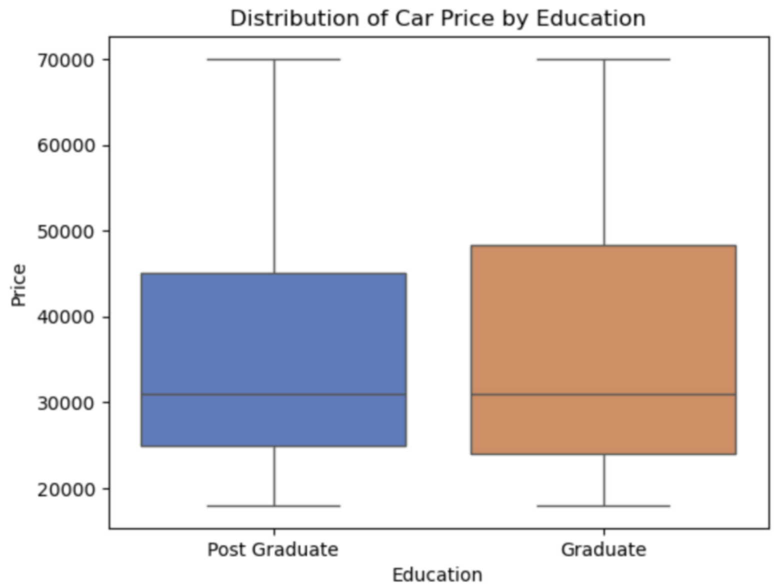
3. Profession



	count	mean	std	min	25%	50%	75%	max
Profession								
Business	685.0	33353.284672	12911.331955	18000.0	23000.0	30000.0	41000.0	70000.0
Salaried	896.0	37313.616071	13926.016563	18000.0	26000.0	33000.0	50000.0	70000.0

Salaried customers spend slightly more than Business Customers and also show a variable price range. Business customers are more concentrated in the ~23-41k Price segment with a few high value outliers indicating that Salaried customers show more varied spending patterns than business customers who are relatively more concentrated in the low-mid price segment.

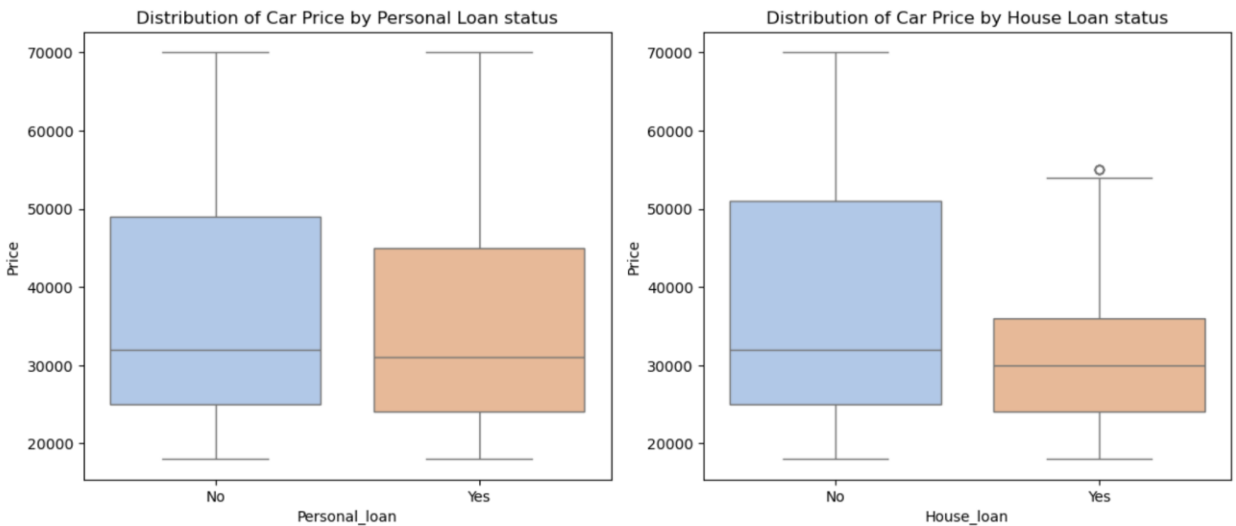
4. Education



Education	count	mean	std	min	25%	50%	75%	max
Graduate	596.0	35907.718121	13997.594301	18000.0	24000.0	31000.0	48250.0	70000.0
Post Graduate	985.0	35410.152284	13412.328181	18000.0	25000.0	31000.0	45000.0	70000.0

Education does not dramatically impact the spending behaviour. However, Graduates tend to show slightly more varied spending compared to Post Graduates.

5. Loan Status



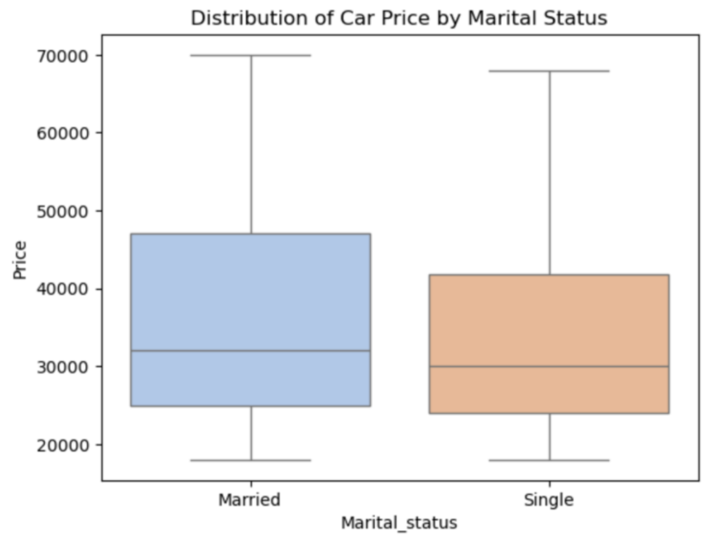
Personal_loan	count	mean	std	min	25%	50%	75%	max
No	789.0	36742.712294	14534.344526	18000.0	25000.0	32000.0	49000.0	70000.0
Yes	792.0	34457.070707	12578.780338	18000.0	24000.0	31000.0	45000.0	70000.0

House_loan	count	mean	std	min	25%	50%	75%	max
No	1054.0	37707.779886	14816.702037	18000.0	25000.0	32000.0	51000.0	70000.0
Yes	527.0	31377.609108	9596.008338	18000.0	24000.0	30000.0	36000.0	55000.0

The distribution of car prices across customers with or without Personal loan is largely similar with nearly identical median and IQR. So, this does not appear to have an impact on spending behavior.

Customers with House loan tend to spend slightly less than the ones without House Loan. Customers without House loan have more varied spending behavior but the core distribution overlaps, and the medians being so close suggest that, loan status is not a primary driver of purchasing behavior.

6. Marital Status

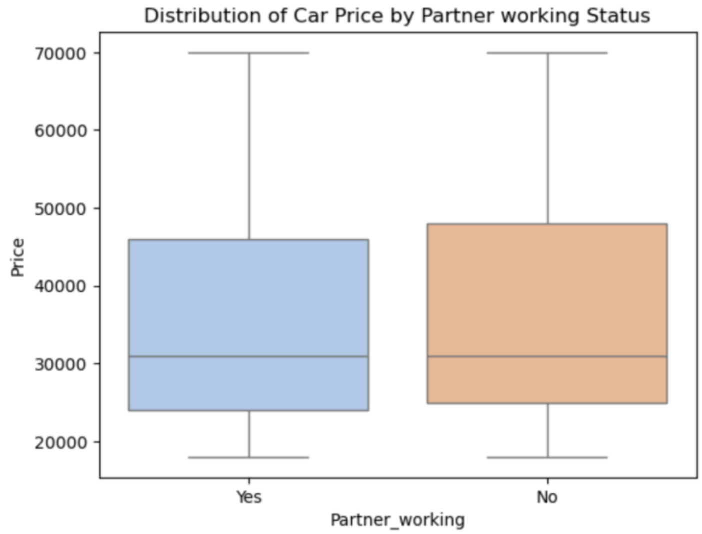


Marital_status	count	mean	std	min	25%	50%	75%	max
Married	1443.0	35800.41580	13723.202856	18000.0	25000.0	32000.0	47000.0	70000.0
Single	138.0	33478.26087	12509.393487	18000.0	24000.0	30000.0	41750.0	68000.0

The distribution of car prices across marital status shows significant overlap, with nearly identical medians for both groups. Married customers show a slightly wider spread in spending, but the overall similarity suggests that Marital status is not a strong driver in purchasing behavior.

Also, due to sample size difference, this dataset is more reflective of a Married person's behavior.

7. Partner Working status

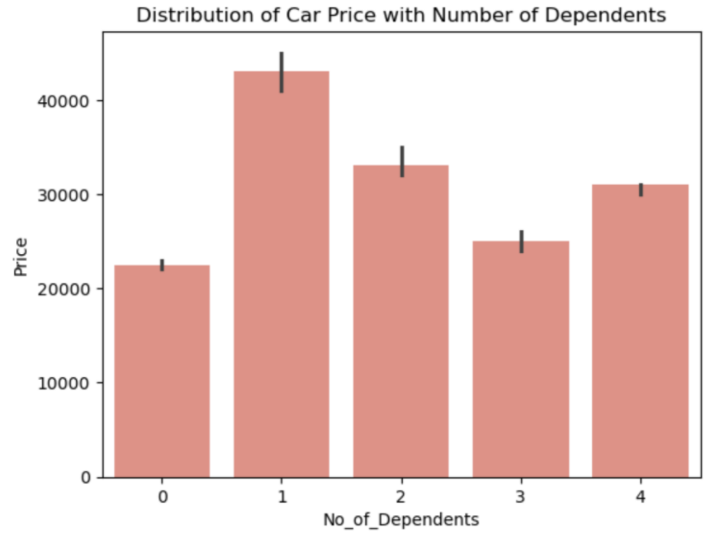


Partner_working	count	mean	std	min	25%	50%	75%	max
No	713.0	36000.000000	13817.734086	18000.0	25000.0	31000.0	48000.0	70000.0
Yes	868.0	35267.281106	13479.532555	18000.0	24000.0	31000.0	46000.0	70000.0

Despite dual-income households having a higher earning potential, the distribution shows a significant overlap with single-income households. This suggests that combined income alone does not automatically translate to purchasing cars from premium segment.

This pattern is further supported by the negative correlation (-0.14) between number of dependents and car prices, suggesting that financial responsibilities may negatively affect spending even in higher earning households.

8. Number of dependents



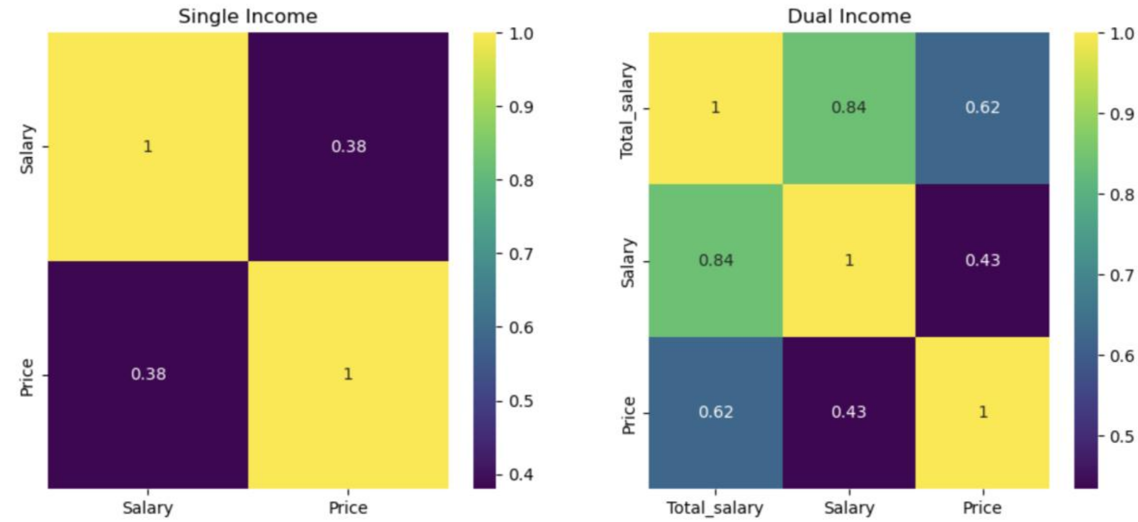
No_of_Dependents	count	mean	std	min	25%	50%	75%	max
0	20.0	28900.000000	13206.457751	18000.0	21750.0	22500.0	29500.0	55000.0
1	229.0	42393.013100	8470.958857	19000.0	35000.0	43000.0	49000.0	69000.0
2	557.0	36615.798923	12633.729333	18000.0	26000.0	33000.0	46000.0	69000.0
3	557.0	31098.743268	14031.924618	18000.0	22000.0	25000.0	32000.0	69000.0
4	218.0	37967.889908	15367.206575	18000.0	29000.0	31000.0	33000.0	70000.0

	No_of_Dependents	Price
No_of_Dependents	1.000000	-0.135839
Price	-0.135839	1.000000

Customers with one dependent show the highest median car price compared to other groups. No_of_Dependents is negatively correlated to Car Price indicating that spending behavior is affected negatively with increase in family responsibility.

Section 3: Income Vs Car Price

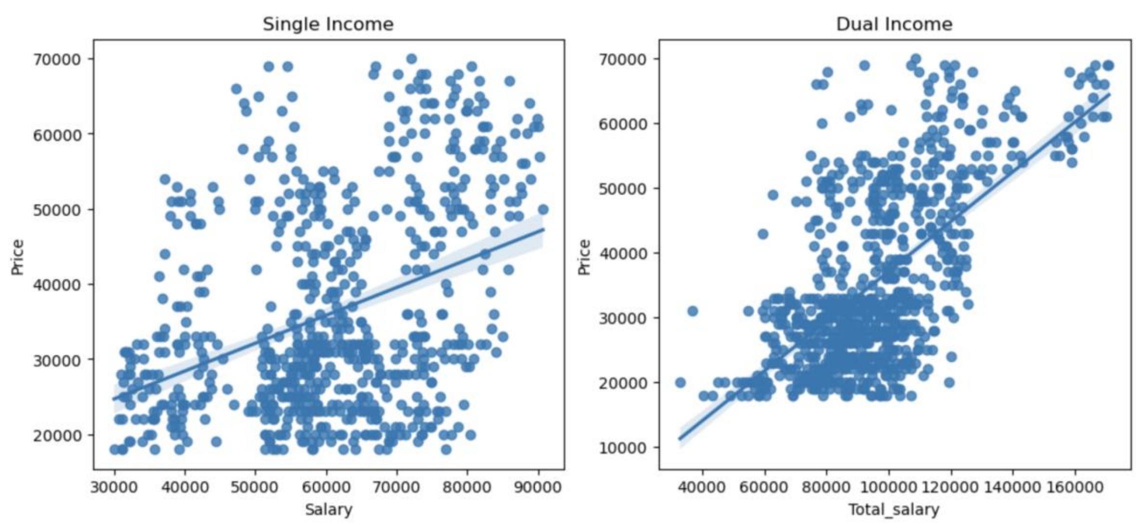
Single income Vs Dual income



For single-income households, the correlation between salary and car price is moderate (0.38), indicating limited influence of individual income on purchasing behavior.

In contrast, for dual-income households, total household income shows a significantly stronger correlation with car price (0.62), compared to individual salary (0.43).

This suggests that purchasing decisions are more closely aligned with overall household earning capacity rather than individual income.



The relationship between income and car price is significantly stronger and more consistent in dual-income households compared to single-income households. While single-income customers show a positive trend, their purchasing behaviour is far more dispersed, indicating the influence of additional factors beyond income. The noticeable spread across income levels in the car price range of 30k suggests that there is a strong preference for mid-segment cars.

```
Partner_working
No     2.305750
Yes    2.582949
Name: No_of_Dependents, dtype: float64
```

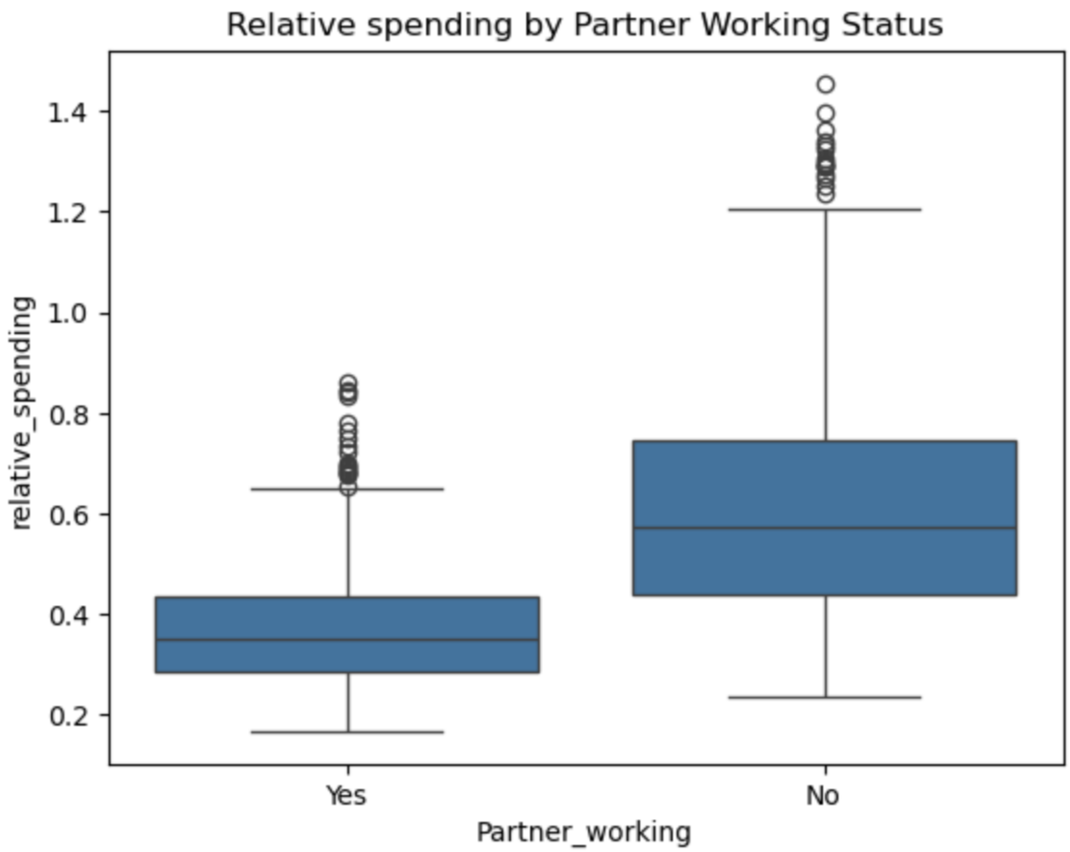
Even though dual income households show a higher earning capacity, they have an average of ~ 2.5 dependents per household, which is slightly higher than that of single income households (~2.3). This suggests that, with increase in responsibilities, the spending capacity is diluted. Therefore, their higher earning capacity does not automatically translate to premium car purchases.

To better understand spending behaviour relative to income, a new variable 'relative_spending' is engineered as the ratio of car price to total household salary.

```
df['relative_spending'] = df['Price']/df['Total_salary']
```

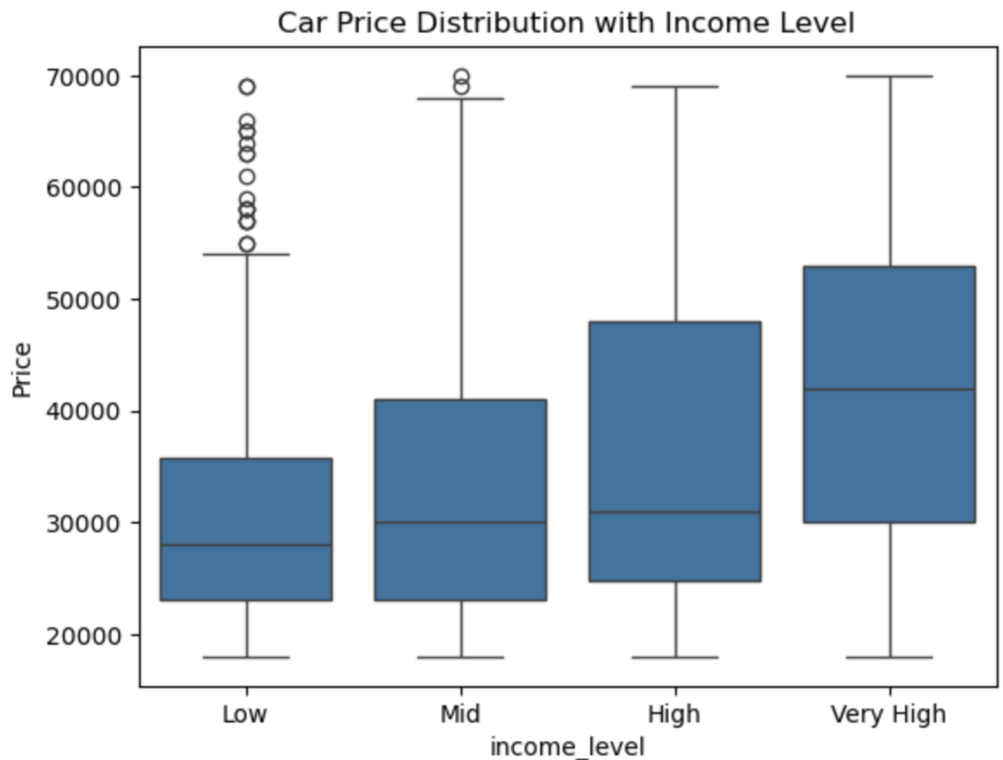
Distribution of Relative Spending by Partner working Status

Partner_working	count	mean	std	min	25%	50%	75%	max
No	713.0	0.612274	0.231739	0.234070	0.437063	0.571429	0.746269	1.455526
Yes	868.0	0.370683	0.114674	0.167224	0.286306	0.351653	0.433290	0.859375



Dual income households have a median relative spending of 0.35 compared to 0.57 for single income households indicating that dual-income customers spend proportionally less of their income on a car. This suggests that dual-income customers are more financially disciplined likely because they have higher average dependents (~2.58 Vs 2.31) and broader financial responsibilities compared to single-income customers.

Another feature called income_level was engineered to understand how Price was distributed among different tiers of Total_salary. This was done by segmenting the Total_salary in quantiles. We have 4 levels – Low, Mid, High and Very high.



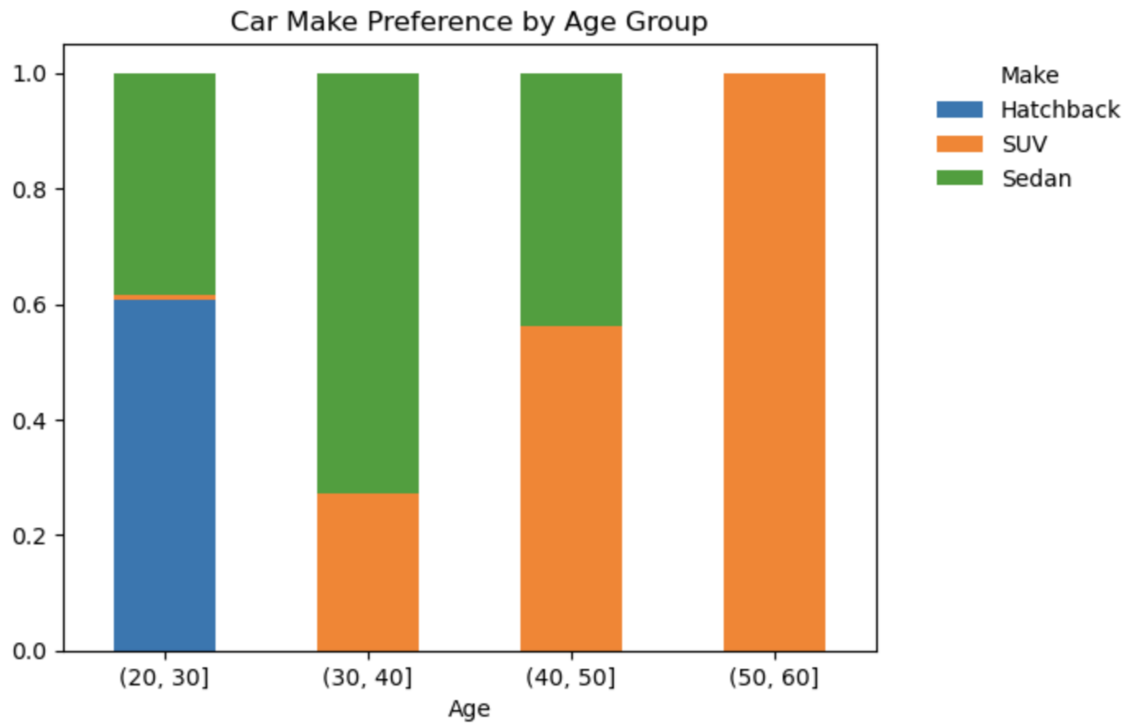
Car prices show a clear upward trend across income levels, with median spending increasing consistently from low to very high income groups. This indicates that household income is a key driver of purchasing power.

However, the presence of high value purchases even within lower income groups suggests that factors such as financing or aspirational buying also influence decisions.

Section 4: Car type Preferences across Segments

1. Age groups

Make Age	Hatchback	SUV	Sedan
(20, 30]	0.607516	0.009395	0.383090
(30, 40]	0.000000	0.272727	0.727273
(40, 50]	0.000000	0.561702	0.438298
(50, 60]	0.000000	1.000000	0.000000

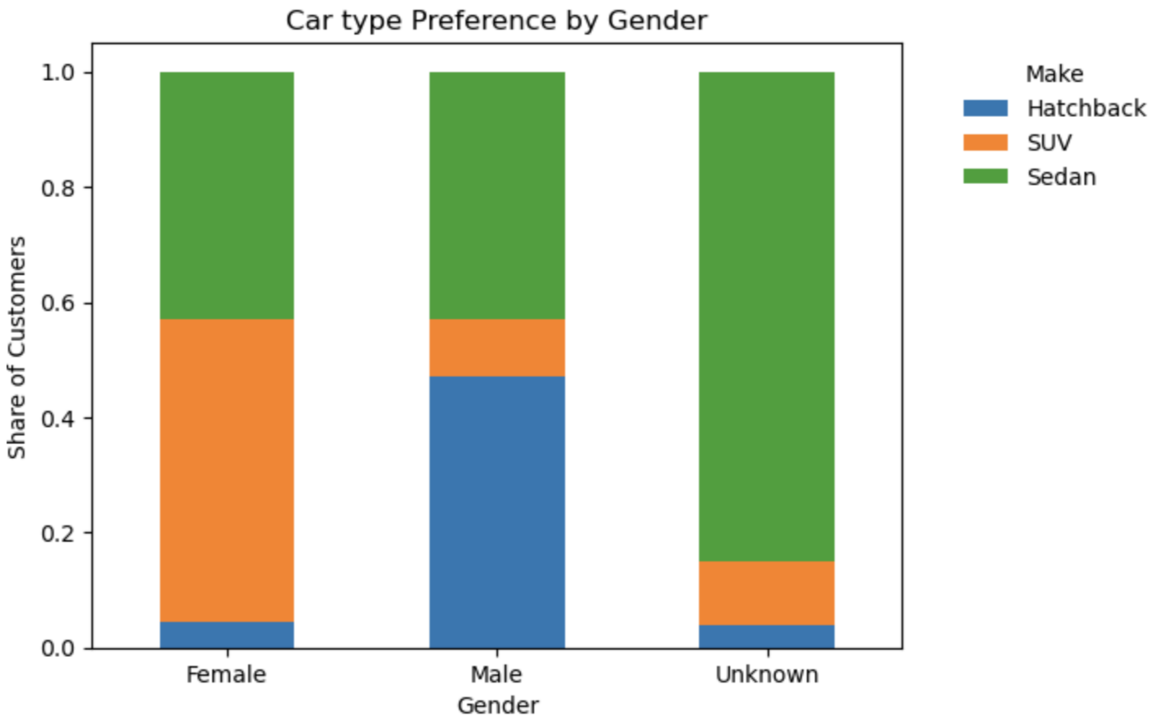


Car preferences evolve with age, with younger customers (20–30) predominantly choosing Hatchbacks, transitioning to Sedans in the 30–40 age group, and showing a higher inclination toward SUVs beyond 40. This indicates a shift from entry-level to premium segments as purchasing power and lifestyle needs evolve.

50-60 age group represents a small sample size, must be interpreted with caution.

2. Gender

Make Gender	Hatchback	SUV	Sedan
Female	0.045593	0.525836	0.428571
Male	0.471226	0.098415	0.430359
Unknown	0.037736	0.113208	0.849057

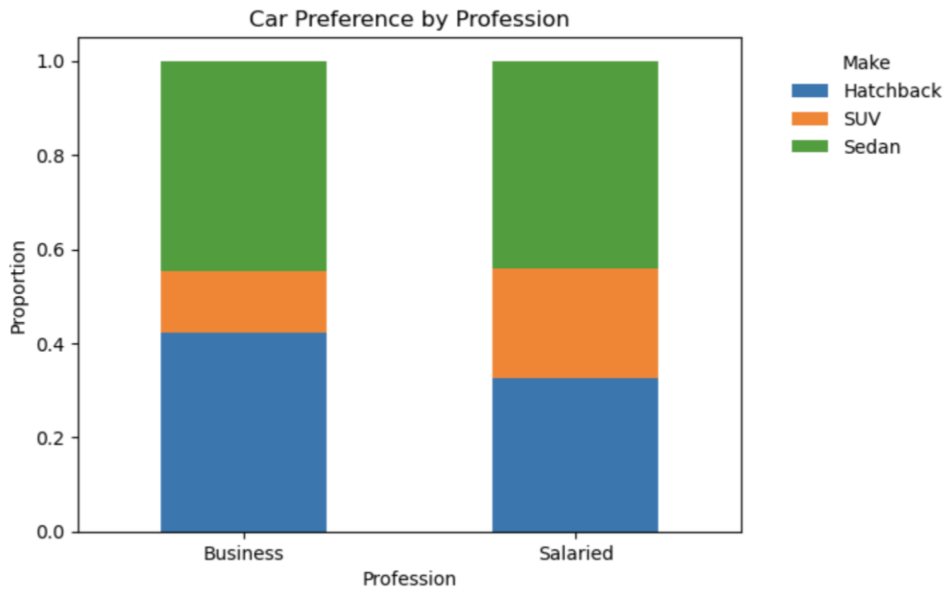


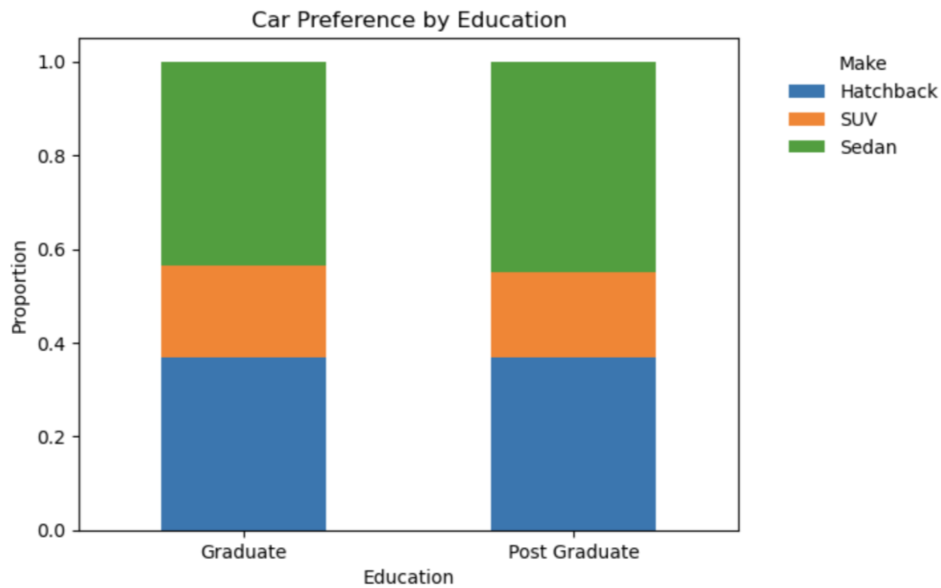
Female customers tend to prefer SUVs more compared to Male customers while Male show a stronger inclination towards Hatchbacks and Sedans. However, the sample size for Female customers is small so interpret with caution.

3. Profession and Education

Make	Hatchback	SUV	Sedan
Profession			
Business	0.423358	0.129927	0.446715
Salaried	0.325893	0.232143	0.441964

Make	Hatchback	SUV	Sedan
Education			
Graduate	0.369128	0.196309	0.434564
Post Graduate	0.367513	0.182741	0.449746



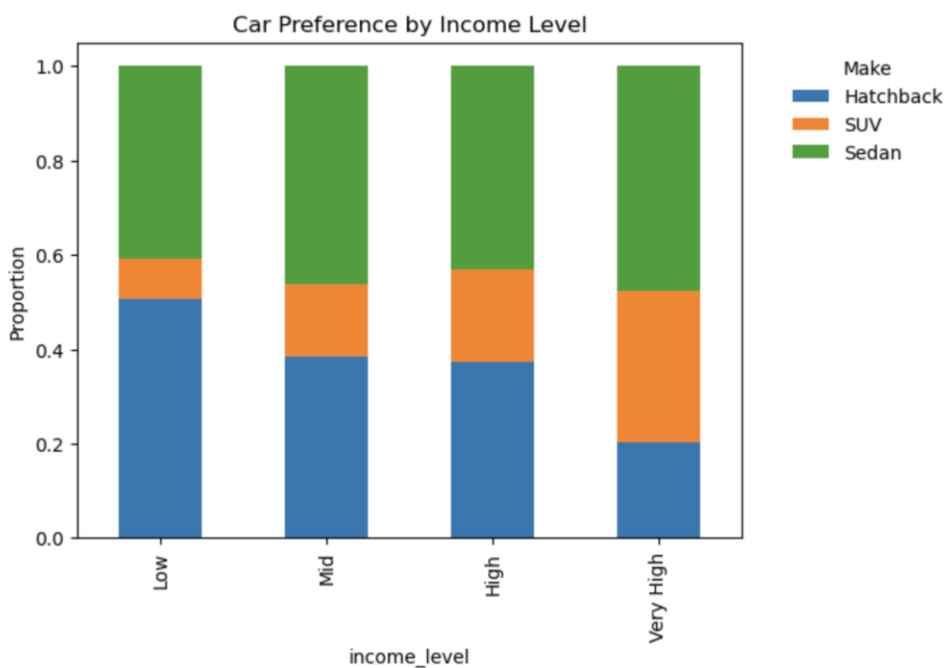


Profession introduces mild variation in car type preference, with business customers leaning toward Hatchbacks and salaried customers toward SUVs, while Sedans maintain consistent popularity across both segments.

Sedan emerges as the most preferred car type across all education levels, indicating its broad appeal across customer segments.

4. Income Level

Make	Hatchback	SUV	Sedan
income_level			
Low	0.507538	0.085427	0.407035
Mid	0.384810	0.154430	0.460759
High	0.373737	0.194444	0.431818
Very High	0.204082	0.318878	0.477041

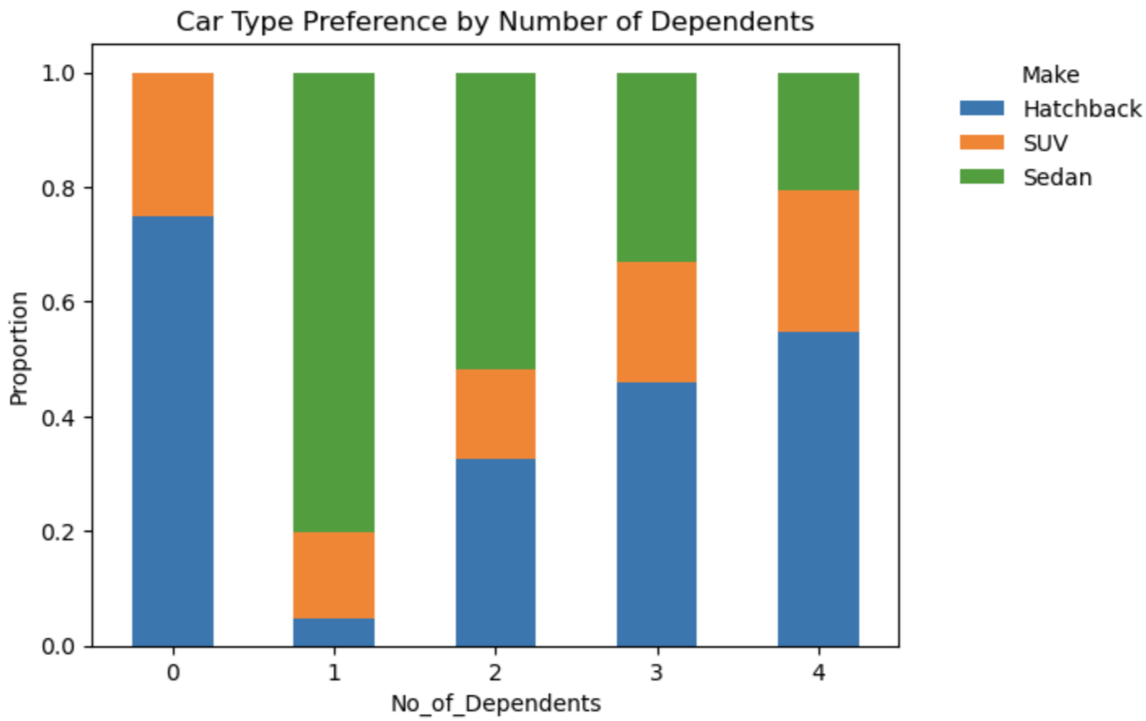


Hatchback preference declines consistently with increasing income, while SUV preference rises, indicating a shift from entry-level to premium segments. Sedan purchases remain relatively stable across all income groups, acting as a mid-segment option that appeals broadly irrespective of income.

Income drives the upgrade path. Sedans continue to retain a strong presence even in higher income groups, indicating that they remain a practical and widely acceptable choice.

4. Number of Dependents

Make	Hatchback	SUV	Sedan
No_of_Dependents			
0	0.750000	0.250000	0.000000
1	0.048035	0.148472	0.803493
2	0.324955	0.156194	0.518851
3	0.459605	0.210054	0.330341
4	0.545872	0.247706	0.206422



SUV preference does not increase with the increase in number of dependents.

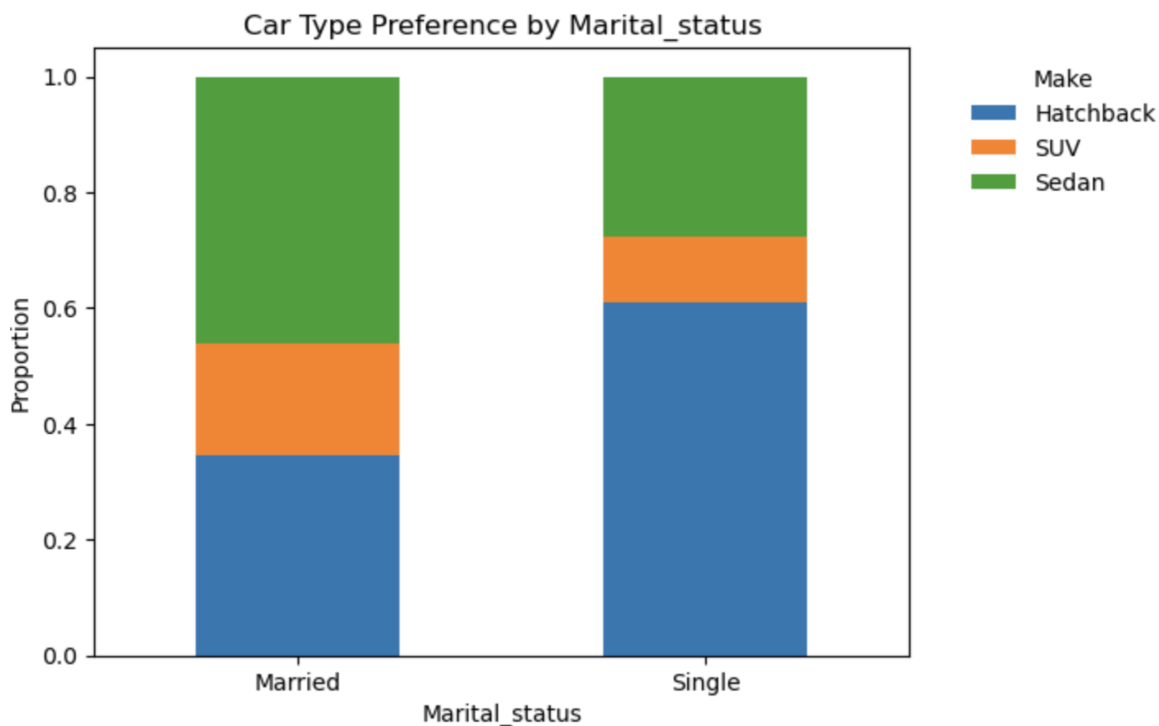
Hatchback Preference peaks at extreme ends of the dependent range.

Sedan is preferred among customers with 1-2 dependents, noticeably dropping when the dependents increase.

This suggests the affordability and financial obligations plays a significant role in the car type

5. Marital Status

Make	Hatchback	SUV	Sedan
Marital_status			
Married	0.345114	0.194733	0.460152
Single	0.608696	0.115942	0.275362

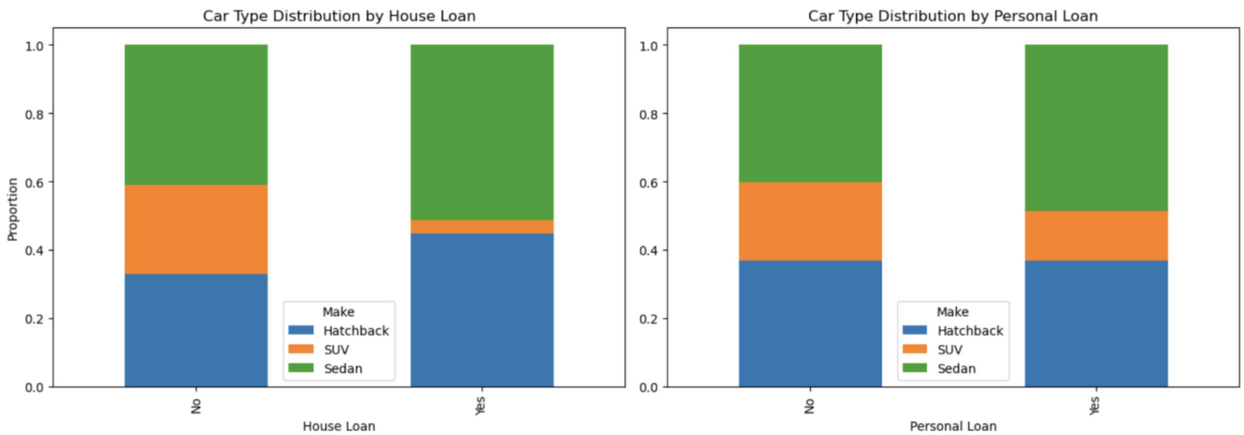


Marital status shows a noticeable difference in car type distribution, with single customers skewed toward Hatchbacks and married customers showing a stronger preference for Sedans. However, since single customers form a small proportion of the dataset, these observations should be interpreted with caution.

6. Loan Status

House_loan	Hatchback	SUV	Sedan
No	0.329222	0.260911	0.409867
Yes	0.445920	0.041746	0.512334

Personal_loan	Hatchback	SUV	Sedan
No	0.368821	0.229404	0.401774
Yes	0.367424	0.146465	0.486111

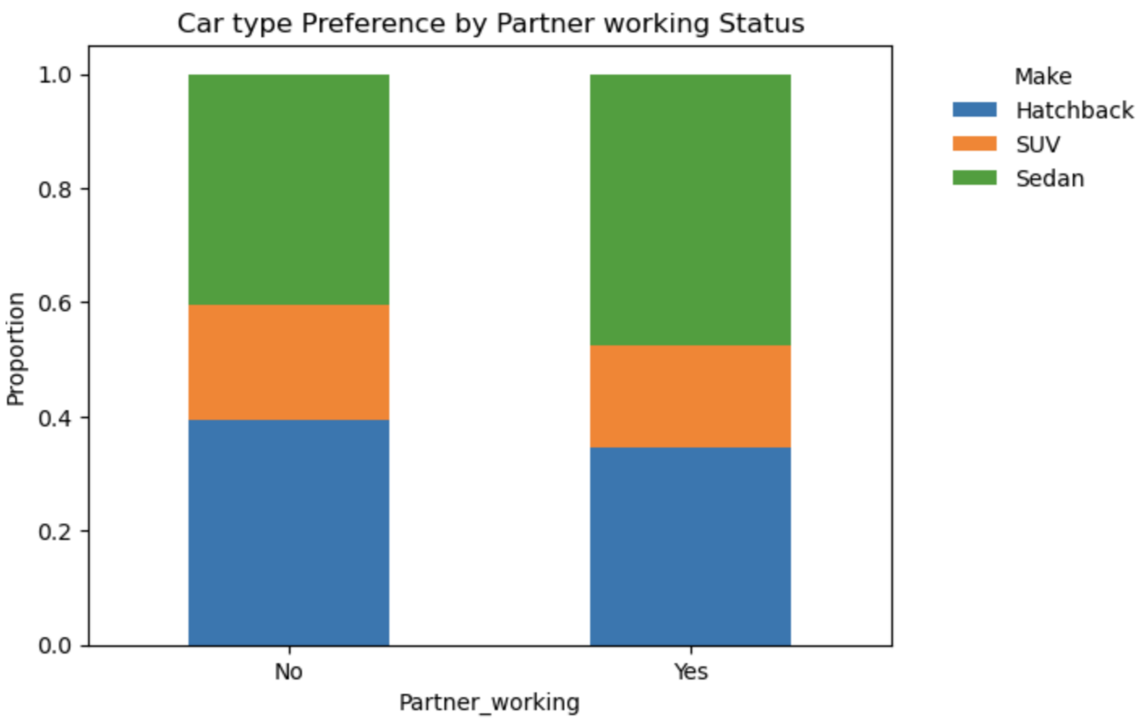


Customers with a house loan show a significantly lower preference for SUVs and a higher inclination toward Hatchbacks and Sedans, indicating the impact of long-term financial commitments on purchase decisions.

In contrast, personal loan status shows only a mild influence, with a slight reduction in SUV preference and increased inclination toward Sedans.

7. Partner Working

Make	Hatchback	SUV	Sedan
Partner_working			
No	0.394109	0.201964	0.403927
Yes	0.346774	0.176267	0.476959



Partner working status shows a mild influence on car type preference, with dual-income households exhibiting a higher inclination toward Sedans, while single-income households are slightly more distributed across Hatchbacks and SUVs.

Take aways – Bivariate Analysis

Customer purchasing behavior in this dataset is primarily driven by age and then income, which together determine the price segment of the car chosen. As income increases, customers move away from Hatchbacks, indicating a clear upgrade path, although the transition toward SUVs is gradual.

Sedans consistently emerge as the most preferred option across income levels, acting as a balanced mid-segment choice. Similarly, with increasing age, customers show a tendency to shift toward higher-priced vehicles, reflecting evolving lifestyle and purchasing capacity.

Gender exhibits a noticeable influence on car type preference, with female customers showing a higher inclination toward SUVs, while male customers predominantly prefer Hatchbacks and Sedans. However, this observation should be interpreted with caution due to differences in sample size.

Financial commitments such as loans do not significantly change overall car type preference but reinforce the dominance of Hatchbacks and Sedans over SUVs. Similarly, variables like partner working status and profession introduce only mild variations, with no strong shift toward premium segments even among dual-income households. Sedans remain consistently preferred across these segments.

5. Critical Business Questions

1. Do men tend to prefer SUVs more compared to women?

Answer: *No, Men do not tend to prefer SUVs more compared to women.*

In this dataset, male customers are more inclined toward Hatchbacks and Sedans, while female customers show a significantly higher preference for SUVs.

Specifically, ~52% of female customers purchase SUVs compared to only ~10% of male customers, indicating that women are substantially more likely to opt for SUVs than men.

Make Gender	Hatchback	SUV	Sedan
Female	0.045593	0.525836	0.428571
Male	0.471226	0.098415	0.430359
Unknown	0.037736	0.113208	0.849057

2. What is the likelihood of a salaried person buying a Sedan?

Answer: The likelihood of a salaried person buying a Sedan is approximately 44%.

This indicates that Sedans are the most preferred car type among Salaried customers, slightly higher than Hatchbacks (~ 32 %) and significantly higher than SUVs (~ 23 %)

This indicates that salaried customers prefer the mid-price segment, which offers a balance between affordability and value, rather than opting for the premium-price segment.

Make Profession	Hatchback	SUV	Sedan
Business	0.423358	0.129927	0.446715
Salaried	0.325893	0.232143	0.441964

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Answer: The data does not support Sheldon Cooper's claim.

Among salaried male customers, only ~13% purchase SUVs, while ~44% opt for Sedans. This indicates that salaried males are significantly more inclined towards Sedans rather than SUVs, making them a weaker target segment for SUV sales.

<u>In essence, the claim is contradicted by data.</u>

	Make	Hatchback	SUV	Sedan
Profession	**Gender**			
Business	**Female**	0.000000	0.523810	0.476190
	Male	0.516995	0.059034	0.423971
	Unknown	0.047619	0.047619	0.904762
Salaried	**Female**	0.066964	0.526786	0.406250
	Male	0.431250	0.132812	0.435937
	Unknown	0.031250	0.156250	0.812500

4. How does the amount spent on purchasing automobiles vary by gender?

Answer: A typical female customer spent substantially higher on car compared to a typical Male customer.

The median price spent by a Female customer is 49k USD while a Male typically spent 29k USD indicating a noticeable difference in spending patterns. However, the Female customer sample size is significantly small compared to Male customers, so this observation should be interpreted with caution.

	count	mean	std	min	25%	50%	75%	max
Gender								
Female	329.0	47705.167173	11244.836378	20000.0	38000.0	49000.0	55000.0	69000.0
Male	1199.0	32817.347790	12299.239195	18000.0	24000.0	29000.0	37500.0	70000.0
Unknown	53.0	23339.622642	10308.591257	18000.0	18000.0	18000.0	19000.0	50000.0

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Answer: Individuals with a personal loan spent a total of 27.3M USD, accounting for 48.5% of the dealership's total revenue.

On an individual basis, they spent an average of ~$31k, which is consistent with cash buyers. This indicates that personal loan status does not significantly influence car purchase spending. Instead, purchasing behavior appears to be driven more by overall affordability than by loan dependency.

```
Amount Spent by individuals with Personal loan:  27290000
Total revenue of Dealership:  56280000
```

Personal_loan	count	mean	std	min	25%	50%	75%	max
No	789.0	36742.712294	14534.344526	18000.0	25000.0	32000.0	49000.0	70000.0
Yes	792.0	34457.070707	12578.780338	18000.0	24000.0	31000.0	45000.0	70000.0

6. How does having a working partner influence the purchase of higher-priced cars?

Answer: Customers with working partners spend proportionally less of their income on cars compared to single income customers.

The median car price between both groups remains identical at ~31K USD indicating that additional income does not translate into purchasing higher priced cars. Also, the relative spending of dual income group is 0.35 and single income group are 0.57. This indicates that customers with working partners are financially more disciplined.

The negative correlation between number of dependents and car Price suggests that with increase in family responsibilities, customer tend to spend rationally. Even though there is a strong correlation between Total_salary and Price of car, it doesn't translate into purchasing premium cars due to such life-stage factors.

Distribution of Car price by Partner working Status

	count	mean	std	min	25%	50%	75%	max
Partner_working								
No	713.0	36000.000000	13817.734086	18000.0	25000.0	31000.0	48000.0	70000.0
Yes	868.0	35267.281106	13479.532555	18000.0	24000.0	31000.0	46000.0	70000.0

Correlation of Total_salary and Car Price in Dual income household

	Total_salary	Price
Total_salary	1.000000	0.623977
Price	0.623977	1.000000

Average no. of dependents per household by partner_working

```
Partner_working
No     2.305750
Yes    2.582949
Name: No_of_Dependents, dtype: float64
```

6. Actionable Insights & Strategic Recommendations

1. Age is the strongest driver of car purchase behavior

   Customers tend to move toward higher-priced vehicles as they age, indicating a shift in preferences and lifestyle.

   - **Recommendation:** Focus on customers aged 40+ for mid-to-premium segments, positioning vehicles around comfort, status, and long-term value rather than just price. Target messaging by life stage, not just income:
     - Younger → affordability, practicality
     - Mid-age → upgrade, family needs
     - Older → comfort, status

2. Gender differences exist but should be interpreted cautiously

   Female customers show a higher median spend compared to male customers. However, this insight is influenced by sample imbalance and should not be overgeneralized.

   - **Recommendation:** Position mid-segment cars as “value for money” to appeal to the dominant male segment, while maintaining aspirational positioning for higher segments without over-targeting based on gender alone.

3. Loan status does not significantly impact spending behavior

   Customers with and without personal loans exhibit similar spending patterns, indicating that financing does not strongly influence the price segment chosen.

   - **Recommendation:** Offer flexible financing and EMI options across all segments rather than targeting specific loan-based customer groups.

4. Sedans are consistently popular among all incomes, professions, and demographics, making them widely appealing.

   Sedans attract both ambitious buyers and those wanting good value.

   - **Recommendation:** Position Sedans as “reliable premium” or “executive comfort” to capture both aspirational and practical buyers.

5. While dual income does not lead to higher spending, there is a slight preference shift toward Sedans.

- **Recommendation:** Target dual-income households with messaging around practicality, comfort, and everyday usability rather than premium positioning alone.

6. Income drives upgrade path across segments Customers transition from Hatchbacks to Sedans (and slightly toward SUVs) as income increases.
   - **Recommendation:** Design tiered marketing:
     - Low income → “affordable, practical” (Hatchback)
     - Mid income → “upgrade, comfort” (Sedan)
     - High income → “status + performance” (SUV/Sedan premium)

7. Conclusion

Customer purchasing behavior at Austo Automobile is primarily driven by age and income, which together determine both the price and segment of cars purchased. As customers progress through life stages and increase their earning capacity, they move away from entry-level hatchbacks toward mid-to-high segment vehicles.

Sedans emerge as the most consistently preferred car type across customer groups, acting as a stable and central segment in the market. While demographic and financial variables such as gender, profession, and loan status introduce some variation, their overall impact on purchasing decisions is limited.

Overall, car purchase behavior is shaped more by lifecycle progression and affordability than by isolated customer attributes, providing clear direction for targeted marketing and product positioning strategies.